

INDU 211/2X 2025 – Assignment #5

Problem 1 (Chapter 15, Problem #1)

Arrivals at a single self-service car wash arrive in a Poisson fashion at a rate of 10 per hour. Service time is exponential with a mean of 4 minutes. If we want to be confident that there is enough room for self-service customers in the system at least 96% of the time, how many cars must we be able to accommodate in the queue?

(Formula: finite geometric series: $\sum_{n=0}^N a^n = \frac{1-a^{N+1}}{1-a}$ ($|a| < 1$))

Problem 2 (Chapter 8, Problem #9 (a))

9. Subgroups of size $n = 5$ are taken from a process. After 40 such subgroups, $\sum_{i=1}^{40} R_i = 930$ and $\sum_{i=1}^{40} \bar{X}_i = 3,000$. The maximum and minimum values of R_i are 45 and 3, respectively. The maximum and minimum values of $\bar{X}_i$ are 86.8 and 63.0, respectively.
- (a) Based on all that you know, is the process in control? Why or why not?

Problem 3 (Chapter 8, Problem #10)

10. An in-control process has $\bar{\bar{X}} = 200$ and $\bar{R} = 46.52$ using subgroups of size $n = 5$.
- (a) What are the $\bar{X} \pm 3\sigma$ values which indicate the natural variability of the process?
- (b) If you would like to have specs at least 5σ from the center of the process, where should they be?

Problem 4 (Chapter 8, Problem #11)

11. Consider a process which, when in control, has $\bar{\bar{X}} = 500$ and $\sigma = 25$.
- (a) Determine the $\bar{X}$ and R control chart limits if $n = 4$.
- (b) Determine the $\bar{X}$ and R control chart limits if $n = 9$.
- (c) Determine the $\bar{X}$ and R control chart limits if $n = 2$.
- (d) Suppose that a shift in process centering occurred from $\bar{\bar{X}} = 500$ to 530. Which subgroup size above yields an $\bar{X}$ chart that is best capable of detecting the shift? Why? What is the major disadvantage of using that sample size?